

# Rajyashree Sen, Ph.D.

## CURRICULUM VITAE

3227 Broadway  
New York, NY 10027  
Tel: 571-271-6764

Email: [rs3966@columbia.edu](mailto:rs3966@columbia.edu)  
Website: <https://rs3966.github.io/>

### EDUCATION

- 2012 – 2017 ● **Ph.D., Molecular Biology & Neuroscience**, *University of Vienna, Austria & Janelia Research Campus, USA*  
Dissertation: Neuronal control of backward walking in *Drosophila melanogaster*  
Advisor: Dr. Barry Dickson
- 2010 – 2012 ● **M.Sc., Biotechnology**, *University of Hyderabad, India*  
Dissertation: Role of Hsp82 in the DNA damage response pathway  
Advisor: Dr. Sunanda Bhattacharyya
- 2007 – 2010 ● **B.Sc., Microbiology**, *St. Xavier's College Kolkata, India*

### PROFESSIONAL EXPERIENCE

- 12/2024 – present ● **Associate Research Scientist**, *Zuckerman Institute of Columbia University*  
A new code for position: flat neural maps in the mouse medial frontal cortex
- 04/2019 – 11/2024 ● **Postdoctoral Research Scientist**, *Zuckerman Institute of Columbia University & Howard Hughes Medical Institute (HHMI)*  
Neuronal basis of individual recognition and social hierarchies in mice  
Advisor: Dr. Richard Axel
- 01/2018 – 03/2019 ● **Postdoctoral Research Associate**, *Janelia Research Campus, USA*  
Distributed control of motor circuits for backward walking in *Drosophila*  
Advisor: Dr. Barry Dickson
- 10/2018 – 12/2018 ● **Invited Visiting Scientist**, *Institute of Zoology at University of Cologne*  
Quantitative analysis of joint kinematics in backward walking flies  
Advisors: Dr. Ansgar Büschges & Dr. Barry Dickson
- 05/2011 – 07/2011 ● **IASc-INSANA Summer Fellow**, *Center for Cellular and Molecular Biology, Hyderabad (CCMB), India*  
Expression and Purification of FAAL10 from *Mycobacterium tuberculosis*  
Advisor: Dr. Rajan Sankarnarayanan

### HONORS

#### RESEARCH

- 07/2025 – 01/2027 ● **Alzheimer's Disease Research Center Research Education Scholar**  
*Alzheimer's Disease Research Center, Taub Institute, Columbia University*
- 01/2025 – 01/2027 ● **BBRF Young Investigator Grant** (\$70,000)  
*Brain and Behavior Research Foundation*
- 10/2012 – 09/2015 ● **Marie Curie Graduate Fellowship**  
*Marie Curie FP7 Programme FLiACT Training Network*
- 08/2010 – 07/2012 ● **Fellowship for Masters in Biotechnology**  
*Full tuition, Department of Biotechnology (DBT), Government of India*
- 05/2011 – 07/2011 ● **Summer Research Fellowship**  
*Indian Academy of Sciences-Indian National Science Academy-The National Academy of Sciences (IASc-INSANA-NASI)*

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| <p><i>PEDAGOGY</i></p> <p>03/2025, 07/2025</p>                                                             | <p>● <b>Teresa Vilardi Scholarship</b><br/><i>Institute of Writing and Thinking, Bard College</i></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <p><i>PUBLIC ENGAGEMENT</i></p> <p>02/2025 – 12/2026</p> <p>01/2025 – 07/2025</p> <p>10/2023 – 05/2024</p> | <p>● <b>The Puffin Foundation Annual Artists Grant</b> (\$2500)<br/><i>The Puffin Foundation Ltd., New Jersey</i></p> <p>● <b>Research! America Civic Engagement Microgrant</b> (\$1800)<br/><i>Research! America, Arlington, Virginia</i></p> <p>● <b>Outreach and Public Engagement Seed Grant</b> (\$750)<br/><i>Zuckerman Institute's Public Engagement Program, Columbia University</i></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|                                                                                                            | <p>Project: <u>The Lab</u>: Bridging Science and Community through Improv Comedy. Used the above grants to conceive, found and produce improv comedy shows inspired by live interviews with neuroscientists. The goal of this project is to engage the general public in scientific discourse.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <p><b>PUBLICATIONS</b></p> <p><i>IN PREPARATION</i></p>                                                    | <p>● <b>Rajyashree Sen</b>, Lorenzo Posani, Alexander Medoff, Larry Abbott, Richard Axel, Stefano Fusi (2026). "A new code for position: Euclidean neural maps in the mouse medial frontal cortex". <i>In Preparation</i>.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <p><i>PREPRINTS</i></p>                                                                                    | <p>● Valentine Andreu, <b>Rajyashree Sen</b>, Nour El Houda Mimouni, Eun Ji Lee, Dianne-Lee Ferguson, Alexis Stutzman, Bianca J. Marlin (2025). "Early postpartum development of pup urine preference in mothers". <i>bioRxiv</i>.<br/>DOI: <a href="https://doi.org/10.1101/2025.09.01.673527">https://doi.org/10.1101/2025.09.01.673527</a></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <p><i>PEER REVIEWED</i></p>                                                                                | <p>● Kai Feng, <b>Rajyashree Sen</b>, Ryo Minegishi, Michael Dübber, Till Bockemühl, Ansgar Büschges, Barry J Dickson (2020). "Distributed control of motor circuits for backward walking in <i>Drosophila</i>" <i>Nature Communications</i>. 11(6166).<br/>DOI: <a href="https://doi.org/10.1038/s41467-020-19936-x">https://doi.org/10.1038/s41467-020-19936-x</a></p> <p>● <b>Rajyashree Sen</b>, Kaiyu Wang, and Barry J. Dickson (2019). "Two lumps ascending neurons mediate touch-evoked reversal of walking direction in <i>Drosophila</i>" <i>Current Biology</i>. 29(24): 4337-4344.e5.<br/>DOI: <a href="https://doi.org/10.1016/j.cub.2019.11.004">https://doi.org/10.1016/j.cub.2019.11.004</a></p> <p>● <b>Rajyashree Sen</b>, Ming Wu, Kristin Branson, Alice Robie, Gerald M. Rubin and Barry J. Dickson (2017). "Moonwalker descending neurons mediate visually evoked retreat in <i>Drosophila</i>" <i>Current Biology</i>. 27(5): 766-771.<br/>DOI: <a href="https://doi.org/10.1016/j.cub.2017.02.008">https://doi.org/10.1016/j.cub.2017.02.008</a></p> |
| <p><b>PRESENTATIONS</b></p> <p>08/2026</p>                                                                 | <p>● <b>Poster</b>   A new code for position: flat neural maps in mouse medial frontal cortex (<b>selected, forthcoming</b>)<br/>Rajyashree Sen, Alexander Medoff, Lorenzo Posani, Larry Abbott, Richard Axel, and Stefano Fusi.<br/><i>Cognitive Computational Neuroscience Conference, New York University, USA</i></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

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| 03/2026                                | <p>● <b>Poster</b>   The dorso-medial prefrontal cortex abstracts space onto a low-dimensional manifold <b>(selected)</b><br/> Rajyashree Sen, Alexander Medoff, Lorenzo Posani, Larry Abbott, Richard Axel, and Stefano Fusi.<br/> <i>Computational and Systems Neuroscience (COSYNE) Conference, Portugal</i></p>                                 |
| 06/2024                                | <p>● <b>Talk</b>   Combined coding of individual identity and spatial locations in the mouse dorso-medial prefrontal cortex (dmPFC) <b>(nominated)</b><br/> <i>Zuckerman Institute Postdoctoral Seminar (ZIPS, Columbia University, USA)</i></p>                                                                                                    |
| 09/2023                                | <p>● <b>Poster</b>   Exposure to different mice elicits different representations in the prefrontal cortex. Rajyashree Sen, Alexander Medoff, Mira Vanchiswar, Larry Abbott, and Richard Axel <b>(contributed)</b><br/> <i>Howard Hughes Medical Institute Science Meeting, USA</i></p>                                                             |
| 11/2020                                | <p>● <b>Featured Speaker</b>   Distributed control of motor circuits for backward walking in <i>Drosophila</i> <b>(invited)</b><br/> <i>Undergraduate Journal Club in Bucknell University, USA (Virtual)</i></p>                                                                                                                                    |
| 08/2019                                | <p>● <b>Talk</b>   Neuronal control of backward walking in <i>Drosophila melanogaster</i> <b>(invited)</b><br/> <i>Indian Institute of Science Education and Research (IISER) Pune, India</i></p>                                                                                                                                                   |
| 05/2018                                | <p>● <b>Talk</b>   Neuronal control of backward walking in <i>Drosophila</i> <b>(nominated)</b><br/> <i>Junior Scientist Seminar Series, Janelia Research Campus, USA</i></p>                                                                                                                                                                       |
| 04/2018                                | <p>● <b>Poster Blitz &amp; Poster</b>   Neuronal control of backward walking in <i>Drosophila melanogaster</i>. <b>(selected)</b><br/> Rajyashree Sen, Till Bockemühl, Kaiyu Wang, Michael Dübbert, Salil Bidaye, Ansgar Büschges and Barry Dickson.<br/> <i>Neural Circuits of the Insect Ventral Nerve Cord, Janelia Research Campus, USA</i></p> |
| 10/2013                                | <p>● <b>Poster</b>   Neural control of directed walking in <i>Drosophila</i>. <b>(selected)</b><br/> Rajyashree Sen, Salil Bidaye and Barry J. Dickson.<br/> <i>Emerging Techniques to Map and Functionally Characterize Neural Circuits in Drosophila, Janelia Research Campus, USA</i></p>                                                        |
| 10/2013                                | <p>● <b>Poster</b>   Neural control of directed walking in <i>Drosophila melanogaster</i>. <b>(selected)</b><br/> Rajyashree Sen, Salil Bidaye and Barry J. Dickson.<br/> <i>Neurobiology of Drosophila Meeting, Cold Spring Harbor Laboratory, USA</i></p>                                                                                         |
| TEACHING                               |                                                                                                                                                                                                                                                                                                                                                     |
| LECTURER<br>Fall 2024                  | <p>● <b>The Parental Brain (Columbia University's Prison Education Project)</b><br/> Adapted and taught a semester-long Columbia University course at <b>Taconic Correctional Facility</b> (medium-security women's prison).</p>                                                                                                                    |
| GUEST LECTURER<br>Fall 2023, Fall 2024 | <p>● <b>The Parental Brain (Seminar)</b><br/> <i>Department of Psychology, Columbia University, USA</i><br/> Developed a module for the course syllabus and led discussions of scientific papers using active learning techniques.</p>                                                                                                              |

## TRAINING

07/2025

### ● Introduction to Writing to Learn in STEM Disciplines

*Institute of Writing and Thinking, Bard College*

Completed a weeklong pedagogical training workshop on using writing as a tool for learning in STEM courses.

03/2025

### ● Teaching for Transfer: Helping Students Remember What They Already Know

*Institute of Writing and Thinking, Bard College*

Completed a one-day pedagogical training workshop on helping students transfer prior knowledge to new contexts.

2017

### ● Scientists Teaching Science (Nine Week Course)

*National Institute of Health*

Completed a comprehensive pedagogical training course designed to strengthen scientists' teaching skills.

## MENTORSHIP

07/2025 – present

### ● Research Assistant: Baihe (Lily) Zhang

02/2025 – present

### ● Undergraduate: Molly Rose Graff (Barnard College)

*Current: Undergraduate at Barnard College, Columbia University*

02/2025 – 07/2025

### ● Undergraduate: Sarafina Belafonte (Columbia University)

*Current: Undergraduate at Columbia University*

01/2023 – 08/2025

### ● Research Assistant: Alexander Medoff

*Current: Graduate student at Columbia University*

04/2024 – 11/2024

### ● Mentor for Black Undergraduate Mentorship Program (BUMP)

05/2023 – 06/2023

### ● Undergraduate: Fatma Albwardy (Columbia University)

*Current: Undergraduate at Columbia University*

01/2020 – 05/2023

### ● Research Assistant: Mira Vanchiswar

*Current: Graduate student at Cornell University*

01/2021 – 05/2023

### ● Undergraduate: Emily Sun (Columbia University)

*Current: Research Assistant at Columbia University*

01/2023 – 04/2023

### ● Undergraduate: Lina Huang (Columbia University)

*Current: Undergraduate at Columbia University*

06/2022 – 08/2023,

### ● Undergraduate: Delaney Bessel (Franklin & Marshall College)

01/2023 – 02/2023

*Current: MS Genetic Counseling Student at Long Island University*

06/2021 – 07/2021

### ● Mentor for Summer Program for Undergraduate Rising Stars (SPURS)

*Supervised Emmanuel Woredokal (Rutgers University)*

*Current: M.D. candidate, Sidney Kimmel Medical College*

01/2018 – 03/2019

### ● Women/Non-Binary Peer Mentoring Group

*Janelia Research Campus, USA*

## TRAINING

2024

### ● Research Mentor Training Workshop

*Center for the Improvement of Mentored Experiences in Research (CIMER)*

Completed a two-day mentorship workshop designed to enhance the mentoring skills of scientists.

## WORKSHOPS

07/2025 – 08/2025

### ● Methods in Computational Neuroscience (selected)

*Marine Biology Laboratory, Woods Hole, USA*

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| 10/2019                  | ● <b>Janelia Junior Scientist Workshop (selected)</b><br>Mechanistic Cognitive Neuroscience<br><i>Janelia Research Campus, USA</i>                                                                                                                                                                   |
| 10/2012 – 09/2015        | ● <b>Marie Curie FLiACT Training Network Workshops on Fly Neuroscience</b><br><i>Max Planck Institute for Chemical Ecology, Jena; Max Planck Institute of Neurobiology, Martinsried; KUL, Leuven, Belgium; Center for Genomic Regulation, Barcelona</i>                                              |
| ACADEMIC SERVICE         |                                                                                                                                                                                                                                                                                                      |
| 01/2025 – present        | ● <b>Columbia Neuroscience Seminar Series – Systems Neuroscience Selection Committee</b><br><i>Columbia University, USA</i>                                                                                                                                                                          |
| 2019 – 2025              | ● <b>Invited Reviewer:</b> eLife, Frontiers in Neuroscience                                                                                                                                                                                                                                          |
| COMMUNICATION & OUTREACH |                                                                                                                                                                                                                                                                                                      |
| IMPROV COMEDY            |                                                                                                                                                                                                                                                                                                      |
| 11/2023 – present        | ● <b>Producer and performer, <i>The Lab</i>:</b> a science-inspired improv comedy show.                                                                                                                                                                                                              |
| 2012 – present           | ● <b>Entertained large audiences</b> for over 10 years as a performer with various improv comedy teams, including house teams at <i>Magnet Theater</i> (New York), <i>Washington Improv Theater</i> (DC), and <i>Laugh Index Theater</i> (DC).                                                       |
| 07/2024 – 07/2025        | ● <b>Second City Conservatory, Second City New York, USA</b>                                                                                                                                                                                                                                         |
| WRITING                  |                                                                                                                                                                                                                                                                                                      |
| 07/2023                  | ● <b>Writing Merit Scholarship – New York State Summer Writers Institute</b><br><i>Skidmore College, New York, USA</i><br>Nominated by Zuckerman Institute's 2022 writer-in-residence Claire Messud.                                                                                                 |
| PUBLIC ENGAGEMENT        |                                                                                                                                                                                                                                                                                                      |
| 2019 – present           | ● <b>Invited Guest Scientist (Science &amp; Public Engagement, New York, NY)</b><br>Featured in a range of NYC science communication events, including <i>Story Collider</i> podcast, the National Jazz Museum's <i>Science and Jazz Talk Show</i> , and miscellaneous science-comedy variety shows. |
| 03/2020                  | ● <b>Finalist, 2020 iBiology Young Scientist Seminars Series Competition</b><br><i>iBiology &amp; Alan Alda Center for Communicating Science</i>                                                                                                                                                     |

| ----- REFERENCES -----                                                                                                                                                                                                           |                                                                                                                                                                                                                  |                                                                                                                                                                                                                    |                                                                                                                                                                                                                               |
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| <b>Richard Axel, M.D.</b><br>(Postdoctoral Advisor)<br>Nobel Laureate, 2004<br>Investigator<br>Zuckerman Institute of<br>Columbia University<br><a href="mailto:ra27@columbia.edu">ra27@columbia.edu</a><br>Tel: +1 212-853-1010 | <b>Stefano Fusi, Ph.D.</b><br>(Postdoctoral Co-advisor)<br>Investigator<br>Zuckerman Institute of<br>Columbia University<br><a href="mailto:sf2237@columbia.edu">sf2237@columbia.edu</a><br>Tel: +1 212-853-1076 | <b>Larry Abbott, Ph.D.</b><br>(Postdoctoral Co-advisor)<br>Investigator<br>Zuckerman Institute of<br>Columbia University<br><a href="mailto:lfa2103@columbia.edu">lfa2103@columbia.edu</a><br>Tel: +1 212-853-1065 | <b>Barry Dickson, Ph.D.</b><br>(Graduate Advisor)<br>Investigator<br>Queensland Brain<br>Institute, The University of<br>Queensland.<br><a href="mailto:b.dickson@uq.edu.au">b.dickson@uq.edu.au</a><br>Tel: +61 7-334 663281 |